



JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal To SRI ABIRAMI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

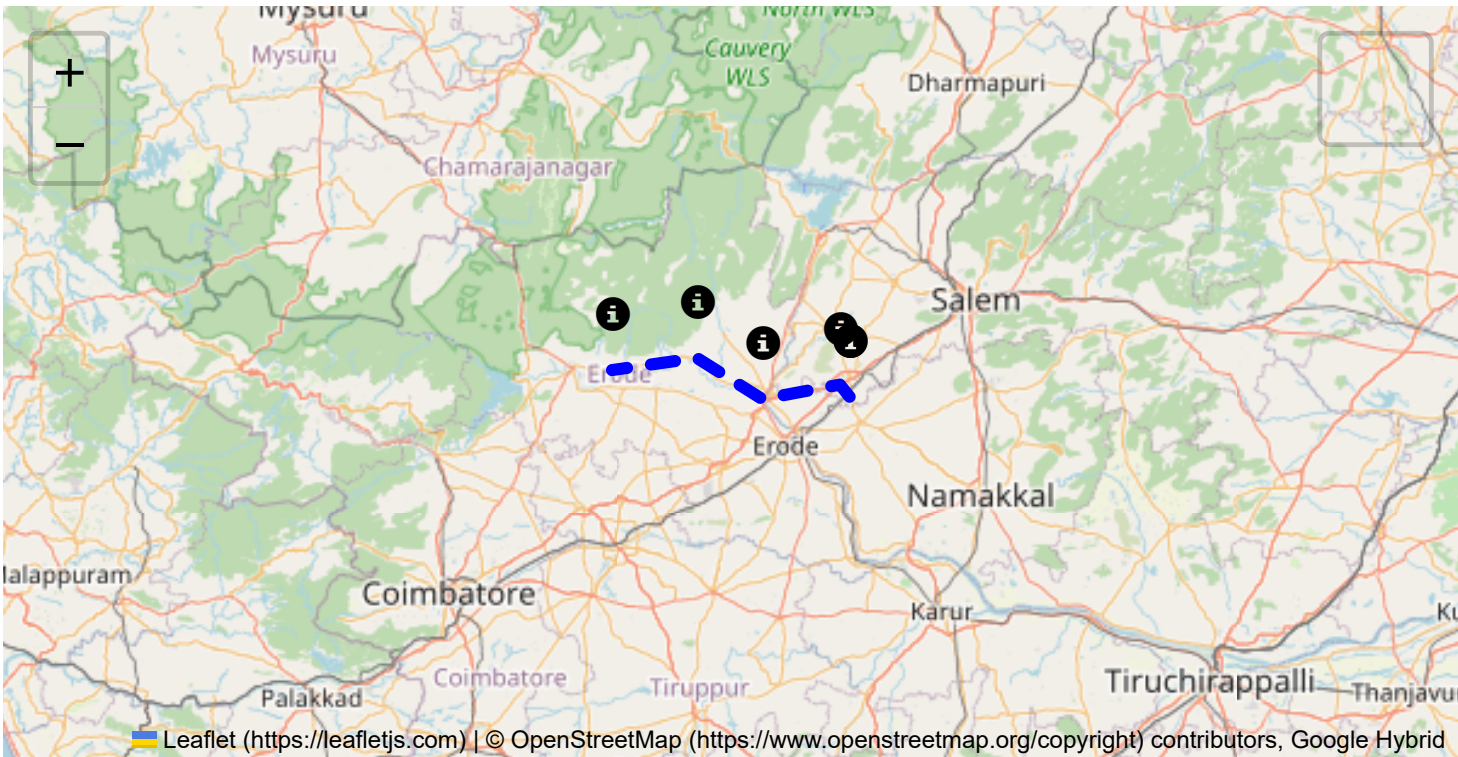
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

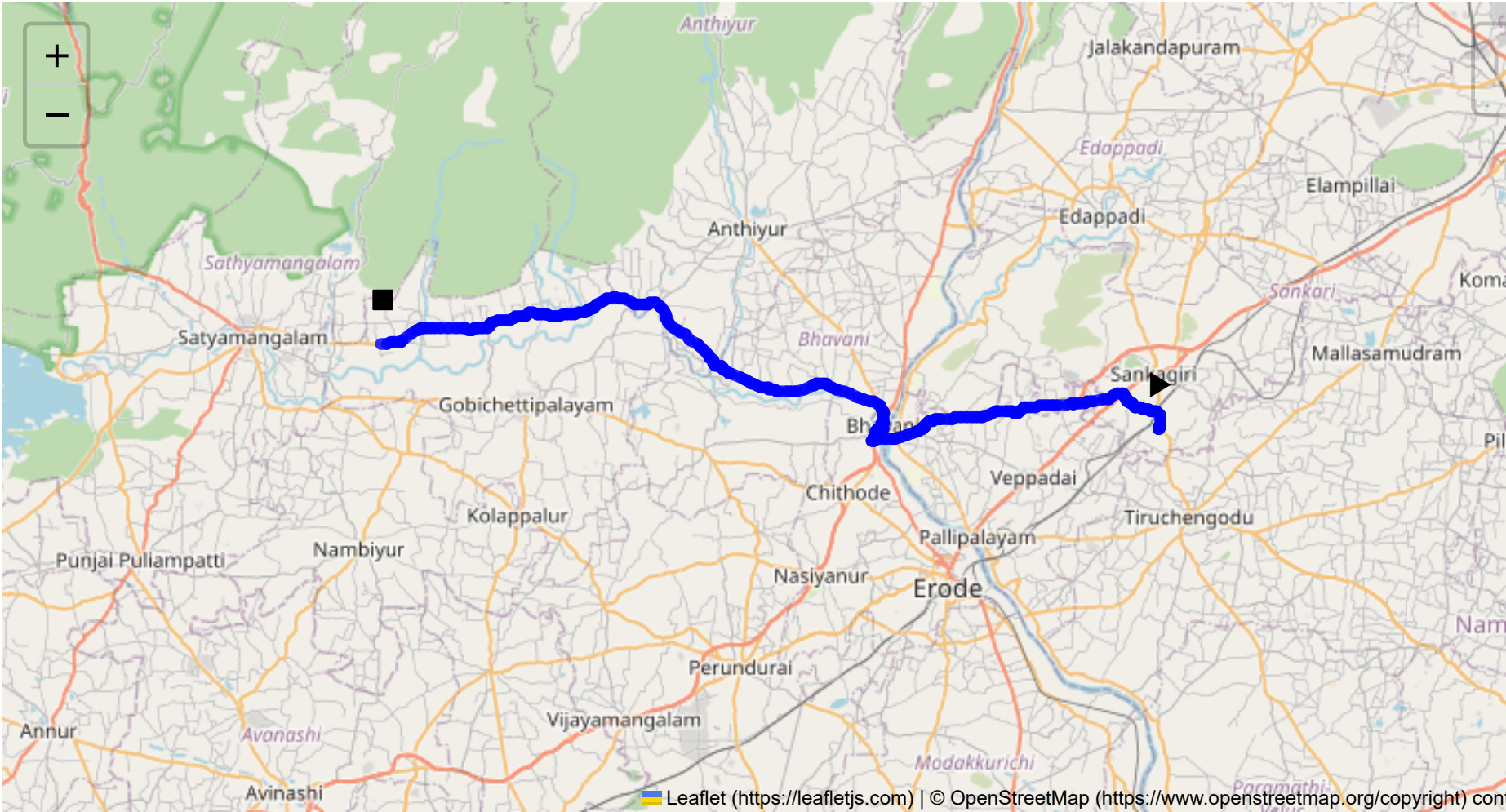
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 70.25 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (11.4964746, 77.3337799)



Welcome to the Journey Risk Management Study

Route Overview

The journey from CVQF+23W, Sangagiri, Tamil Nadu to F8WM+HG Thookkanaickenpalayam, Tamil Nadu, is approximately 70.25 kilometers, primarily following state highways and main roads. This route is used for transporting goods and involves several rural and urban segments.

Typical Weather Conditions

The region experiences a tropical climate with hot summers and moderate monsoons. **Weather-related Hazards:**

- **Summer (March to June):** High temperatures can affect vehicle efficiency.
- **Monsoon (July to September):** Roads may be slippery; potential for flooding and reduced visibility.

Traffic Patterns

Traffic tends to peak during morning (7:00-9:00 AM) and evening (5:00-7:00 PM) hours, particularly in areas like Sangagiri and Bhavani. The SH 79A and Bhavani Main Road are prone to congestion, especially near markets and junctions.

Road Quality and Infrastructure

While the state highways (like SH 79A) are generally well-maintained, some interior roads may suffer from potholes or are narrower, which could affect larger vehicles. Regularly updated traffic apps can help avoid trouble spots.

Alternative Routes for Emergencies

In case of roadblocks or major delays, consider:

- Diverting through Erode when traveling north.
- Utilizing the NH 544 at Sangagiri for a different route alignment, though it might slightly increase the distance.

Local Regulations on Hazardous Material Transport

The state of Tamil Nadu mandates licenses for transporting hazardous materials, requiring adherence to specific safety protocols and routes. Vehicles must display appropriate placards and follow speed limits strictly, particularly in populated areas.

Historical Incidents

Historically, there have been several minor accidents possibly due to vehicular failure or road conditions. However, records of significant incidents involving hazardous materials are limited, though perpetual vigilance is essential.

Environmental Considerations

- The route passes through agricultural areas and minor forested zones. Ensure adherence to noise and emission norms to minimize impact.
- Proper disposal of waste and leaks from transport vehicles is crucial to protect land and water in these areas.

Communication Coverage

Telecom coverage is typically good in urban and semi-urban sections but can be spotty in rural depths, particularly around forested or hilly tracts. It is advisable to have updated offline maps and a vehicle communication system.

Emergency Response Times

Estimated emergency services response times vary between:

- **15-20 minutes** in urban centers.
- **30-45 minutes** in rural or less accessible areas. Availability of medical assistance and vehicle repair services decreases significantly as you move away from major towns like Bhavani.

Overall Summary of Risk Assessment

While the route is generally manageable for truck drivers, especially those transporting hazardous materials, several risk factors demand attention:

- 1. **Weather and Road Conditions:** Monitor forecasts and road advisories.
- 2. **Traffic and Infrastructure:** Plan to avoid peak congestion times.
- 3. **Regulatory Compliance:** Strict adherence to transportation laws and safety standards for hazardous materials is non-negotiable.
- 4. **Preparedness:** Maintain a backup communication plan and familiarize with alternative routes.

Ensuring that drivers are well-trained and vehicles are properly equipped will mitigate most risks encountered along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44893, 77.87413	15 KM/Hr
4	Turn	Medium	11.45127, 77.86734	30 KM/Hr
5	Turn	Medium	11.45354, 77.85798	30 KM/Hr
6	Turn	High	11.45350, 77.85700	15 KM/Hr
7	Turn	Medium	11.45845, 77.85141	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.46317, 77.84906	30 KM/Hr
10	Turn	High	11.43000, 77.67564	15 KM/Hr
11	Turn	Medium	11.43009, 77.67540	30 KM/Hr
12	Turn	High	11.43026, 77.67525	15 KM/Hr
13	Turn	High	11.43776, 77.67816	15 KM/Hr
14	Turn	High	11.43864, 77.68225	15 KM/Hr
15	Turn	Medium	11.43965, 77.68224	30 KM/Hr
16	Turn	High	11.44105, 77.68188	15 KM/Hr
17	Turn	High	11.44120, 77.68218	15 KM/Hr
18	Turn	High	11.44288, 77.68254	15 KM/Hr
19	Turn	High	11.44298, 77.68273	15 KM/Hr
20	Turn	High	11.44858, 77.68333	15 KM/Hr
21	Turn	Medium	11.48131, 77.57024	30 KM/Hr
22	Turn	Medium	11.48204, 77.56636	30 KM/Hr
23	Turn	Medium	11.48529, 77.56464	30 KM/Hr
24	Turn	Medium	11.52406, 77.52552	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.52745, 77.50262	30 KM/Hr
26	Turn	Medium	11.51441, 77.45439	30 KM/Hr
27	Turn	Medium	11.50784, 77.40840	30 KM/Hr
28	Turn	Medium	11.50506, 77.39651	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital	11.4525596, 77.7749426	30 km/h	Medium
4	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
5	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
6	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
7	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
9	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
10	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
11	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
12	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
13	hospital	Sutha Hospital	11.4365526, 77.6796472	30 km/h	Medium
14	hospital	Maha Hospital	11.4372003, 77.6781345	30 km/h	Medium
15	hospital	Jothikumar Hospital	11.43687, 77.678407	30 km/h	Medium
16	hospital	SKMH Hospital	11.4379787, 77.6780268	30 km/h	Medium
17	clinic	Sri Dharshan Diabetes Care	11.437649, 77.6778126	30 km/h	Medium
18	clinic	Velavan Homoeo Clinic	11.4367087, 77.6761472	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	hospital	Kavin Hospital	11.4399114, 77.6838194	30 km/h	Medium
20	clinic	Thangam Acupuncture	11.4418816, 77.6832318	30 km/h	Medium
21	hospital	Annusiya Hospital	11.4432777, 77.6813562	30 km/h	Medium
22	hospital	Sumangali Hospital	11.4427668, 77.6827131	30 km/h	Medium
23	hospital	Kandasamy Hospital	11.4450062, 77.6832304	30 km/h	Medium
24	hospital	Senthil Hospital	11.4475854, 77.6844482	30 km/h	Medium
25	hospital	S.N.A. Siddha Hospital	11.44704, 77.682877	30 km/h	Medium
26	hospital	Ashok Krishna Hospital	11.4483344, 77.6856579	30 km/h	Medium
27	hospital	Shri Mani Hospital	11.44771, 77.6838471	30 km/h	Medium
28	clinic	Sri Hari Clinic	11.4483503, 77.6850963	30 km/h	Medium
29	hospital	Immiaditya Hospital	11.4501276, 77.6836088	30 km/h	Medium
30	hospital	Sri Saravana Hospital	11.4506202, 77.6829676	30 km/h	Medium
31	hospital	Government Hospital	11.45112, 77.682149	30 km/h	Medium
32	hospital	Vijay Hospital	11.45091, 77.682019	30 km/h	Medium
33	hospital	Anbe Sivam Hospital	11.4508139, 77.6854947	30 km/h	Medium
34	clinic	Kalyani Nursing Home	11.452097, 77.6808974	30 km/h	Medium
35	hospital	Government Hospital	11.4519113, 77.6819742	30 km/h	Medium
36	hospital	Sriram Child and Maternity Hospital	11.4739074, 77.5783099	30 km/h	Medium
37	hospital	NM Hospital	11.4765419, 77.5801178	30 km/h	Medium
38	hospital	Dr. Kandhasamy Hospital	11.4750159, 77.5798168	30 km/h	Medium
39	hospital	SPK Hospital	11.47498, 77.5798076	30 km/h	Medium
40	clinic	Krithik Clinic	11.4756073, 77.5780846	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
41	hospital	Vmk Hospital	11.4791501, 77.5724052	30 km/h	Medium
42	hospital	Subi Hospital	11.5070348, 77.5375885	30 km/h	Medium
44	hospital	Government Hospital	11.5218505, 77.5285238	30 km/h	Medium
45	hospital	Kariyamanikam Hospital	11.5229297, 77.5134812	30 km/h	Medium
46	hospital	Bala Hospital	11.522996, 77.512794	30 km/h	Medium
47	hospital	Nalam General & Diabetes Care Hospital	11.523399, 77.51107	30 km/h	Medium
49	hospital	KRP Hospital	11.5142473, 77.4546226	30 km/h	Medium
50	pharmacy	Sumi Medicals	11.5157965, 77.4506732	30 km/h	Medium
51	hospital	Government Hospital, T.N.Palayam	11.50771, 77.383285	30 km/h	Medium
52	hospital	Hussainya Homeopathy Hospital	11.507225, 77.386729	30 km/h	Medium
53	hospital	Karthikeyan Hospital	11.5068171, 77.3844296	30 km/h	Medium
54	hospital	Dr. P Kathirvel Hospital	11.5064531, 77.3836952	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
8	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
43	marketplace	Weekly Market	11.5078119, 77.5364552	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
48	school	BEST Matriculation High School	11.5173252, 77.4633376	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44893, 77.87413



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45127, 77.86734



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45354, 77.85798



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45350, 77.85700



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45845, 77.85141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46317, 77.84906



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43000, 77.67564



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43009, 77.67540



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43026, 77.67525



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43776, 77.67816

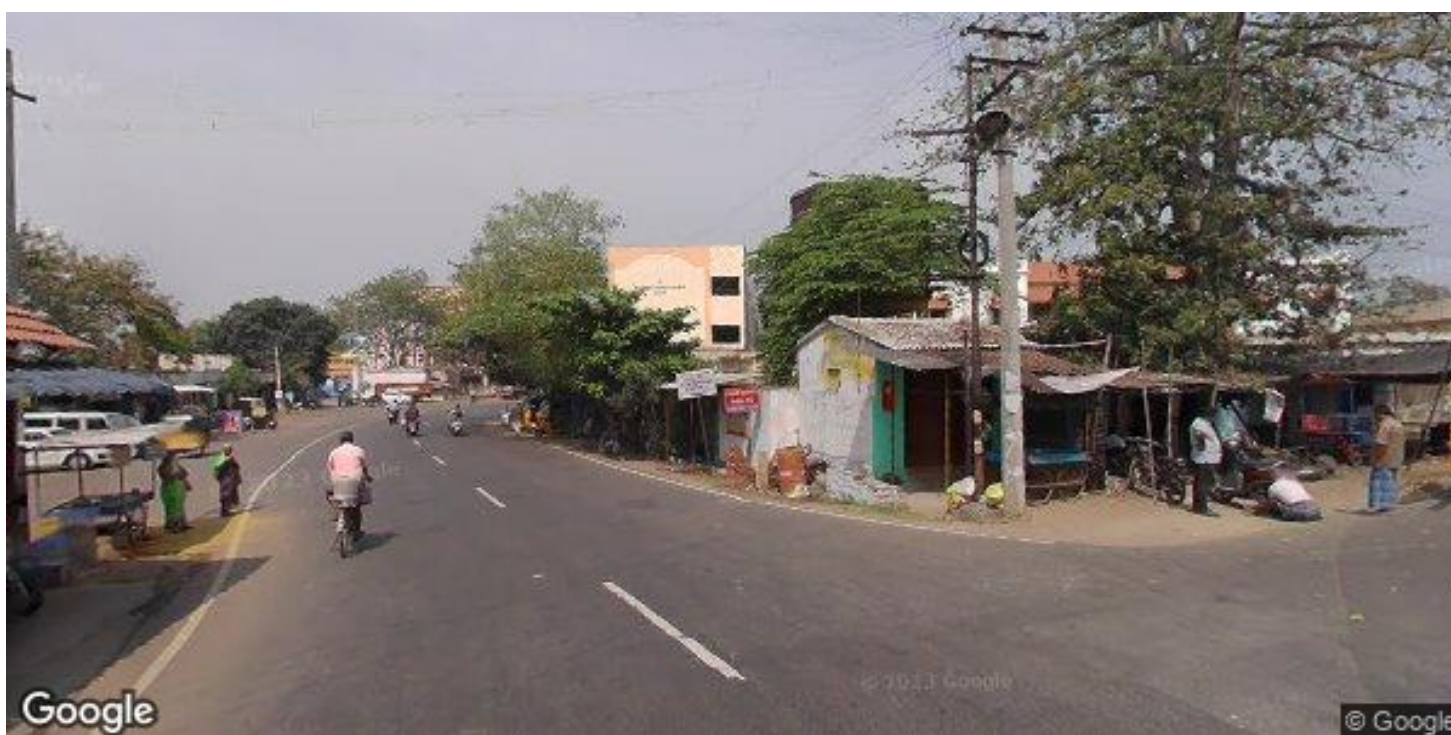


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43864, 77.68225



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43965, 77.68224



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44105, 77.68188



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44120, 77.68218



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44288, 77.68254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44298, 77.68273



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44858, 77.68333



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48131, 77.57024



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48204, 77.56636



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48529, 77.56464



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52406, 77.52552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52745, 77.50262



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.51441, 77.45439



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50784, 77.40840



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50506, 77.39651
